

H.S. 2nd Year — Computer Science and Applications (2024)

Solved Question Paper — Code: 34T CSCA (Bengali)

Full Marks: 70 | Pass Marks: 21 | Time: Three hours

Q. No. 1 — Answer any eight ($1 \times 8 = 8$) [All parts solved]

(a) What is C++ ?

Ans: C++ is a general-purpose, object-oriented, middle-level programming language developed by Bjarne Stroustrup at Bell Labs as an extension of the C language. It supports both procedural and object-oriented programming.

(b) What is an object ?

Ans: An object is a basic run-time entity of object-oriented programming. It is an instance of a class that contains data members (attributes) and member functions (methods) which operate on that data. Example: a variable of a class type.

(c) What is class ?

Ans: A class is a user-defined data type that acts as a blueprint or template for creating objects. It binds data and functions together (encapsulation) and defines the properties and behaviour that its objects will have.

(d) What is inheritance ?

Ans: Inheritance is an OOP feature by which a new class (derived class) acquires the properties and behaviour (data members and member functions) of an existing class (base class). It promotes code reusability.

(e) What is concrete class ?

Ans: A concrete class is a class that has complete implementation of all its member functions and can be used to create (instantiate) objects directly. It is the opposite of an abstract class, which cannot be instantiated.

(f) What are data members ?

Ans: Data members are the variables declared inside a class that hold the data/attributes of the class. They represent the state of an object, e.g., in a class 'Student', roll_no and name are data members.

(g) What is a data structure ?

Ans: A data structure is a systematic way of organising, storing and managing data in a computer so that it can be accessed and modified efficiently. Examples: array, stack, queue, linked list, tree, graph.

(h) Give examples of non linear data structure.

Ans: Examples of non-linear data structures: Tree and Graph. (In these, data elements are not arranged sequentially; each element may connect to more than one other element.)

(i) What is sequential allocation ?

Ans: Sequential allocation (also called static/contiguous allocation) is a method of storing data elements in consecutive (contiguous) memory locations, as done in arrays, so that elements can be accessed directly using an index.

(j) What is LIFO ?

Ans: LIFO stands for 'Last In First Out'. It is the principle followed by a Stack, in which the element inserted last is the first one to be removed/deleted.

(k) Give two examples of header files used in C++.

Ans: Two examples of C++ header files: `<iostream>` (for input/output operations like cin, cout) and `<conio.h>` (for console I/O functions like getch(), clrscr()). Other examples: `<string.h>`, `<math.h>`.

(l) What is database ?

Ans: A database is an organised collection of related data stored electronically in a computer system, which can be easily accessed, managed, and updated. It is managed using a Database Management System (DBMS), e.g., MySQL.

(m) What is LAN ?

Ans: LAN stands for Local Area Network. It is a network that connects computers and devices within a small geographical area, such as a single building, office, or school/college campus.

Q. No. 2 — Answer any ten ($2 \times 10 = 20$) [All parts solved]

(a) Define polymorphism.

Ans: Polymorphism means 'many forms'. It is an OOP feature that allows the same function name or operator to behave differently based on the context, e.g., a function 'area()' calculating area differently for a circle and a rectangle. It is of two types: compile-time (function/operator overloading) and run-time (function overriding).

(b) State the advantages of object oriented programming languages.

Ans: Advantages: (1) Code reusability through inheritance, (2) Data security through encapsulation and data hiding, (3) Easy maintenance and modification of code, (4) Reduces code complexity through polymorphism, (5) Real-world problems can be modelled easily using objects.

(c) Define member function.

Ans: A member function is a function that is declared and defined inside a class (or its scope) and operates on the data members of that class. It defines the behaviour of the objects of that class, e.g., `getdata()` and `putdata()`.

(d) Define constructor.

Ans: A constructor is a special member function of a class that has the same name as the class and no return type. It is automatically invoked whenever an object of the class is created, and is mainly used to initialise data members.

(e) Define pointer.

Ans: A pointer is a variable that stores the memory address of another variable, instead of storing a data value directly. It is declared using the '*' symbol, e.g., `int *p;` where `p` holds the address of an integer variable.

(f) Define non-primitive data structure.

Ans: A non-primitive data structure is a structure that is derived from primitive data types and can store a collection of values, either of the same type or different types. Examples: array, list, stack, queue, tree, graph, file.

(g) Define two-dimensional array.

Ans: A two-dimensional array is an array of arrays, arranged in the form of rows and columns (like a matrix/table), used to store data in a tabular format. It is declared as: `data_type array_name[row_size][column_size];` e.g., `int a[3][4];`

(h) Give examples of DBMS.

Ans: Examples of DBMS software: MySQL, Oracle, Microsoft SQL Server, Microsoft Access, PostgreSQL, MongoDB (NoSQL DBMS).

(i) State the categories of SQL.

Ans: SQL commands are categorised into: (1) DDL (Data Definition Language) – e.g. CREATE, ALTER, DROP; (2) DML (Data Manipulation Language) – e.g. INSERT, UPDATE, DELETE; (3) DQL (Data Query Language) – e.g. SELECT; (4) DCL (Data Control Language) – e.g. GRANT, REVOKE; (5) TCL (Transaction Control Language) – e.g. COMMIT, ROLLBACK.

(j) State two uses of Boolean algebra.

Ans: Two uses of Boolean algebra: (1) It is used to design and simplify digital logic circuits used in computers and calculators. (2) It is used in the design of switching circuits and simplification of complex logical expressions.

(k) What is the use of Truth table ?

Ans: A truth table is used to show all the possible combinations of input values of a Boolean/logical expression along with the corresponding output values, helping to verify and analyse the behaviour of logic gates and circuits.

(l) What is POS in Boolean algebra ?

Ans: POS stands for 'Product of Sums'. It is a form of expressing a Boolean expression as an AND (product) of two or more OR (sum) terms, e.g., $(A + B).(A' + C)$.

(m) What is Bandwidth ?

Ans: Bandwidth is the maximum rate of data transfer across a network or communication channel in a given amount of time, usually measured in bits per second (bps). Higher bandwidth means more data can be transmitted per second.

(n) What is FTP ?

Ans: FTP stands for File Transfer Protocol. It is a standard network protocol used to transfer files (upload/download) between a client and a server over a TCP/IP-based network such as the Internet.

Q. No. 3 — Answer any nine ($3 \times 9 = 27$) [All parts solved]

(a) Explain the concept of function overloading.

Ans: Function overloading is a feature of C++ in which two or more functions have the same name but differ in the number or type of their parameters. The compiler decides which function to call based on the arguments passed, at compile time. It is a type of compile-time polymorphism. Example: `void add(int a,int b);` and `void add(float a,float b);` — both named 'add' but overloaded.

(b) Briefly describe the concept of base class and derived class.

Ans: Base class (or parent class) is the existing class whose properties and methods are inherited by another class. Derived class (or child class) is the new class that inherits the data members and member functions of the base class and can also add its own new members or redefine existing ones. This relationship is established through inheritance, e.g., `'class Student : public Person'` — here Person is the base class and Student is the derived class.

(c) Describe about dynamic memory allocation.

Ans: Dynamic memory allocation is the process of allocating memory to variables during the run-time (execution) of a program, rather than at compile time. In C++, this is done using the 'new' operator (to allocate memory) and the 'delete' operator (to free/release memory). It allows a program to use memory efficiently, since memory is allocated only when needed and can be released after use. Example: `int *p = new int; ... delete p;`

(d) Compare static data structure with dynamic data structure.

Ans: Static data structure: (1) Memory is allocated at compile time. (2) Size is fixed and cannot be changed during execution. (3) Example: Array. Dynamic data structure: (1) Memory is allocated at run time. (2) Size can grow or shrink during execution. (3) Example: Linked List. Thus, static structures are simple but inflexible, while dynamic structures use memory efficiently but need pointers to manage links.

(e) Write the difference between primary key and candidate key.

Ans: Primary key: (1) It is the candidate key that is selected by the database designer to uniquely identify each record in a table. (2) A table can have only ONE primary key. (3) It cannot contain NULL values. Candidate key: (1) It is any column or set of columns that qualifies to uniquely identify records (i.e., is a potential primary key). (2) A table can have MULTIPLE candidate keys. (3) Candidate keys other than the chosen primary key are called alternate keys.

(f) State the applications of Karnaugh Map.

Ans: Applications of Karnaugh Map (K-Map): (1) Used to simplify Boolean/logical expressions without using lengthy Boolean algebra laws. (2) Used in the design of digital logic circuits with the minimum number of logic gates. (3) Helps to easily identify redundant terms in a Boolean expression. (4) Used to minimise SOP (Sum of Products) and POS (Product of Sums) expressions, reducing hardware cost and circuit complexity.

(g) Give examples of different data transfer rates.

Ans: Different units of data transfer rate (examples): (1) bps – bits per second, (2) Kbps – Kilobits per second (1,000 bps), (3) Mbps – Megabits per second (1,000 Kbps), (4) Gbps – Gigabits per second (1,000 Mbps). For example, a broadband connection may offer a speed of 100 Mbps, while fibre-optic networks may offer speeds in Gbps.

(h) Differentiate WAN and MAN.

Ans: WAN (Wide Area Network): (1) Covers a very large geographical area such as a country or across countries. (2) Ownership is usually with multiple organisations/public. (3) Example: The Internet. MAN (Metropolitan Area Network): (1) Covers a city or a town, larger than a LAN but smaller than a WAN. (2) Usually connects several LANs within a city. (3) Example: Cable TV network of a city.

(i) Write three properties of XML.

Ans: Three properties of XML (eXtensible Markup Language): (1) XML is extensible – users can define their own custom tags as per requirement, unlike HTML which has predefined tags. (2) XML is platform-independent and self-descriptive – the tags describe the meaning of the data they contain. (3) XML separates data from presentation, making it ideal for data storage and exchange between different systems/applications.

(j) State the differences between Worm and Trojan horse.

Ans: Worm: (1) It is a self-replicating malicious program that spreads automatically from one computer to another over a network without needing a host file. (2) It does not need to attach itself to another program. Trojan horse: (1) It is a malicious program disguised as a legitimate/useful software to trick users into installing it. (2) It does not replicate itself but relies on the user to execute it, and it can create backdoors for hackers.

(k) Compare 3G with 4G technologies.

Ans: 3G (Third Generation): (1) Provides data speed typically in the range of 384 Kbps to a few Mbps. (2) Supports basic video calling and mobile internet browsing. 4G (Fourth Generation): (1) Provides much higher data speed (up to 100 Mbps or more), based on IP technology (LTE). (2) Supports high-quality video streaming, faster browsing and better multimedia services compared to 3G.

(l) State the uses of different aggregate functions in SQL.

Ans: Aggregate functions in SQL perform a calculation on a set of values and return a single value. Common ones: (1) SUM() – calculates the total sum of a numeric column. (2) AVG() – calculates the average value of a numeric column. (3) COUNT() – counts the number of rows/records. (4) MAX() – returns the maximum value in a column. (5) MIN() – returns the minimum value in a column. These are widely used with GROUP BY for summarising data.

(m) Describe the operations applicable to Queue.

Ans: The main operations applicable to a Queue (a FIFO – First In First Out – data structure) are: (1) Enqueue (Insertion) – adding an element at the rear end of the queue. (2) Dequeue (Deletion) – removing an element from the front end of the queue. (3) Peek/Front – viewing the element at the front without removing it. (4) isEmpty() – checking whether the queue is empty. (5) isFull() – checking whether the queue is full (in case of an array-based queue).

Q. No. 4 — Answer any three ($5 \times 3 = 15$) [All parts solved]

(a) Explain the differences between Stack and Queue.

Ans: Stack: (1) Follows LIFO (Last In First Out) principle. (2) Insertion and deletion take place from the same end, called the 'top'. (3) Operations: push() (insert) and pop() (delete). (4) Example use: undo operation in editors, function call management (recursion). Queue: (1) Follows FIFO (First In First Out) principle. (2) Insertion takes place at the 'rear' end and deletion takes place at the 'front' end — i.e., two different ends. (3) Operations: enqueue()

(insert) and dequeue() (delete). (4) Example use: CPU scheduling, printer queue management. Thus, the fundamental difference lies in the order of removal of elements and the ends used for insertion/deletion.

(b) Write a C++ program to create a matrix of order 4×4.

Program (accepts elements from user and displays the 4×4 matrix):

```
#include <iostream>
using namespace std;
int main()
{
    int a[4][4], i, j;
    cout << "Enter elements of 4x4 matrix:" << endl;
    for (i = 0; i < 4; i++)
    {
        for (j = 0; j < 4; j++)
        {
            cout << "Element [" << i << "][" << j << "]: ";
            cin >> a[i][j];
        }
    }
    cout << "\nThe 4x4 Matrix is:\n";
    for (i = 0; i < 4; i++)
    {
        for (j = 0; j < 4; j++)
        {
            cout << a[i][j] << "\t";
        }
        cout << endl;
    }
    return 0;
}
```

(c) Explain three different network topologies.

Ans: Network topology refers to the physical/logical arrangement of computers and devices in a network. Three common topologies: (1) Bus topology – All devices are connected to a single central cable called the 'bus' or 'backbone'. It is simple and cost-effective for small networks, but if the main cable fails, the whole network goes down. (2) Star topology – All devices are connected individually to a central device (hub/switch). It is easy to install and manage, and failure of one node does not affect others; however, if the central hub fails, the whole network fails. (3) Ring topology – Each device is connected to exactly two other devices, forming a circular path for data (token passing). Data travels in one direction (or both), and it performs well under heavy load, but failure of one node/cable can break the entire ring (unless a dual ring is used).

(d) Explain the concept of Cyber Law and Cyber Crimes.

Ans: Cyber Law refers to the legal framework/set of rules that deals with the legal issues related to the use of the Internet, computers, and cyberspace — including electronic transactions, digital signatures, data protection, and online activities. In India, it is governed mainly by the Information Technology (IT) Act, 2000. Cyber Crime refers to any illegal/criminal activity that is carried out using a computer, computer network, or the Internet as a tool or target. Examples of cyber crimes include hacking, phishing, identity theft, spreading viruses/malware, online fraud, cyber stalking, and data theft. Cyber laws are formulated precisely to prevent, detect, and punish such cyber crimes and to protect individuals' and organisations' rights in cyberspace.

(e) State De Morgan's law and explain the applications of it.

Ans: De Morgan's Law consists of two theorems in Boolean algebra: (1) $(A + B)' = A'.B'$ — The complement of a sum (OR) equals the product (AND) of the complements. (2) $(A.B)' = A' + B'$ — The complement of a product (AND) equals the sum (OR) of the complements. Applications: De Morgan's laws are widely used to simplify complex Boolean/logical expressions and digital circuits. They allow conversion between AND, OR and NOT

operations (e.g., converting an AND-OR circuit into an equivalent NOR/NAND-only circuit), which is especially useful in circuit design since NAND and NOR gates are 'universal gates' and cheaper to manufacture. They also help in minimising logic circuits, thus reducing the number of gates and overall cost.

(f) Explain the advantages of C++ programming language over other programming languages.

Ans: Advantages of C++ over other programming languages: (1) It supports Object-Oriented Programming (OOP) features like encapsulation, inheritance, and polymorphism, which make code more organised, reusable and secure — unlike purely procedural languages such as C. (2) It is a middle-level language, combining the features of both high-level and low-level languages, so it supports both system programming and application programming. (3) It offers faster execution speed compared to many other high-level languages, as it is a compiled language close to hardware. (4) It provides rich library support (STL - Standard Template Library) for data structures and algorithms. (5) It supports both procedural and object-oriented paradigms, giving programmers flexibility. (6) It allows direct memory manipulation using pointers, giving greater control over system resources than languages like Java or Python.